

Cover letter

Dear Editor,

Please consider my manuscript, **Shrinkage linear replicate supervision recovers perturbation identity and pharmacological relationships across LINCS contexts**, for publication as a methods and benchmarking article.

The study addresses a practical question in perturbational transcriptomics: when replicate identity supervises a retrieval representation, does a neural metric learner or a classical shrinkage estimator transfer more reliably? The core idea of replicate-supervised L1000 representation learning was established by Filzen and colleagues in 2017, which I cite and reconstruct directly. My contribution is a stricter compound-disjoint and no-retraining transfer benchmark that identifies a different practical conclusion.

Using GSE92742 for training and GSE70138 for frozen transfer, shrinkage linear discriminant analysis outperformed raw expression, PCA, limma weighting, Hallmark ssGSEA, a modern supervised-contrastive encoder, and a faithful reconstruction of the published perturbation-barcode network. The result held in five new cell lines comprising 390 unseen compound-contexts and 2,593 plate-level profiles. The same frozen geometry also improved retrieval of shared mechanisms of action and protein targets between distinct unseen compounds. Training and held-out generalized covariance analyses, together with a 20-point scaling grid, were consistent with a sample-efficiency advantage for shrinkage LDA. A longer-schedule audit improved both neural comparators but left LDA first at the full 100-compound point and throughout the 20-compound row.

The manuscript is deliberately bounded. It does not claim invention of replicate supervision, therapeutic equivalence, platform-independent validation, or prospective biological confirmation. It presents an honest benchmark showing that a well-regularized classical model can be more transferable than the tested one-hidden-layer neural alternatives at a scale of approximately 100 training compounds.

All data are public. The submission package contains frozen analysis contracts, exact compound splits, executable code, compact derived data, model checkpoints, query-level results, and artifact hashes. The 25-artifact core replay and a separate two-artifact reviewer-response replay were both byte exact. The work received no external funding, and the author declares no competing interests.

Thank you for your consideration.

Sincerely,

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